

**Course:** BTech**Semester:** 1

Prerequisite: Students should possess basic computer literacy and familiarity with using web-based applications. Basic communication skills and logical thinking are recommended for effective interaction with AI tools. No prior knowledge of programming or Artificial Intelligence is required.

Rationale : This course introduces the fundamentals of Prompt Engineering to enable effective interaction with AI systems. It helps students develop AI-assisted problem-solving, creativity, and productivity skills for academic and professional applications.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	-	-	-	3	20	20	-	60	-	100

SEE - Semester End Examination, T - Theory, P - Practical

Course Content

W - Weightage (%) , T - Teaching hours

Sr.	Topics	W	T
1	Introduction to LLM (Generative AI) and Prompt Engineering: Introduction to AI, Machine Learning vs Deep Learning vs Generative AI, Introduction to Prompt Engineering, History of LLM and Chat GPT and Claude, Application of Prompt Engineering, Opportunity and Career Scope, Explore and Compare Different LLMs (ChatGPT, Claude, Google Gemini, Grok AI, DeepSeek).	15	7
2	Fundamental of Prompt Writing: Structure of a Prompt (Role, Task, Context, Constraints, Output Format), Prompt Type (Open-ended, Close-ended, Instruction Prompt, Question Prompt, Persona Prompt), Prompt Templet, Writing Good Prompt, Create Prompt for Essay, Email, Resume, Letter, Story, Blog)	15	7
3	Prompt Engineering Technique: Component of Effective Prompting, Zero-shot, One-shot, few-shot, Step by Step, Self Reflection, Role-Based, Persona, Output Control (JSON, Table, Bullet Point, Structure), Prompt Optimization, Create Prompt for Travel Planer, Architecture, Study Note, Code Generation, Mathematics, Summarization.	20	8
4	Advance Prompt Engineering Technique: Prompt Chaining, Strategic Role Player, ReAct, iterative(self) Refinement, Multi-step and hierarchical Prompt, Prompt decomposition, classify customer feedback into one of three categories: [Pricing, Support, Features], You are a financial analyst. Calculate the YoY growth for our 2026 revenue, Create Code for specific Task.	20	8
5	AI Productivity tools and Multimodal Prompting: Text AI, Image Generation, Presentation Generation, Document AI, Spreadsheet AI, Video AI, Speech AI, Introduction to Multimodal, Use AI tools to create Images, Posters, Logos, Presentation, Infographics, Audio Transcript, Video Script.	15	7
6	Mini Project and Project Demonstration: Creating Website and AI Model using Prompt Engineering	15	8
Total		100	45

Reference Books

1.	The AI Prompt Engineering Handbook, (TextBook) By Jules White Vanderbilt University
2.	Prompt Engineering for LLMs: The Art and Science of Building Large Language Model-Based Applications By John Berryman and Albert Ziegler O'Reilly Media, Pub. Year 2024
3.	OpenAI Prompt Engineering Guide By OpenAI OpenAI Documentation
4.	Prompt Engineering Interactive Tutorial and Best Practices By Anthropic Anthropic Documentation



Course Outcomes

At the end of this course Students Will be able to:

1	Explain the fundamentals of Generative AI and Prompt Engineering.
2	Design effective prompts for different AI tasks.
3	Apply prompt engineering techniques to solve academic and professional problems.
4	Evaluate AI responses considering ethics, bias, privacy, and reliability.
5	Develop simple AI-powered solutions using prompt engineering and present a mini project.